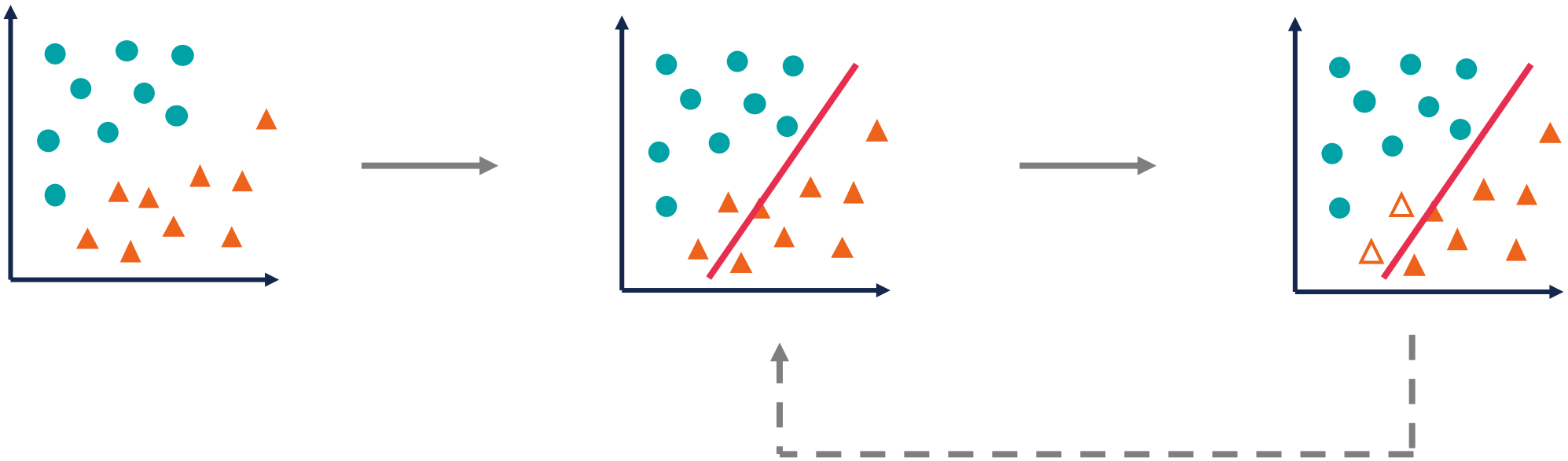
02 Data Preprocessing

DATA

MODEL

Learns patterns
& decision rules

PREDICTION



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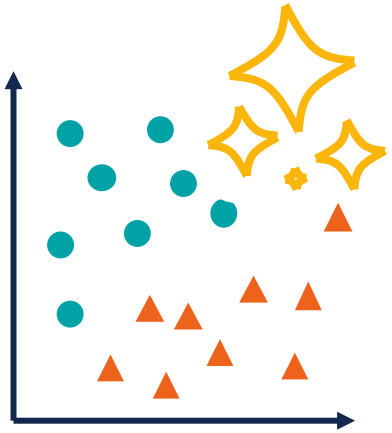
Garbage *in* Garbage **OUT**

https://medium.com/nyc-design/gigo-garbage-in-garbage-out-concept-for-ux-research-7e3f50695b82

Data pre-processing

CLEAN DATA



Missing data

Different scales

Non-numerical values

Wrong values

Missing data

Eccentricity	Diameter (mm)	Color	Border
0.1	4	No	NaN
0.3	3	No	Yes
0.9	6	Yes	NaN
0.3	2	Yes	NaN
NaN	3	No	NaN
0.6	NaN	No	NaN

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- Remove rows/columns with many missing values

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- Remove rows/columns with many missing values
- Remove rows with important missing values

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- Remove rows/columns with many missing values
- Remove rows with important missing values
- Imputation
  - Numeric → mean/median
  - Categorical → most frequent value
  - K-nearest neighbors
  - Model-based (e.g. regression)

clean data vs sample size

Imputations are also ”impure”

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Data normalization

Eccentricity	Diameter (mm)	Color
0.1	4	No
0.3	3	No
0.9	6	Yes
0.3	2	No
0.3	3	No

Features with large ranges can dominate model training

→ biased weight updates
→ slow or unstable learning

Min-max scaling $$x_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}}$$

Standardization $$x_{norm} = \frac{x - \mu}{\sigma}$$

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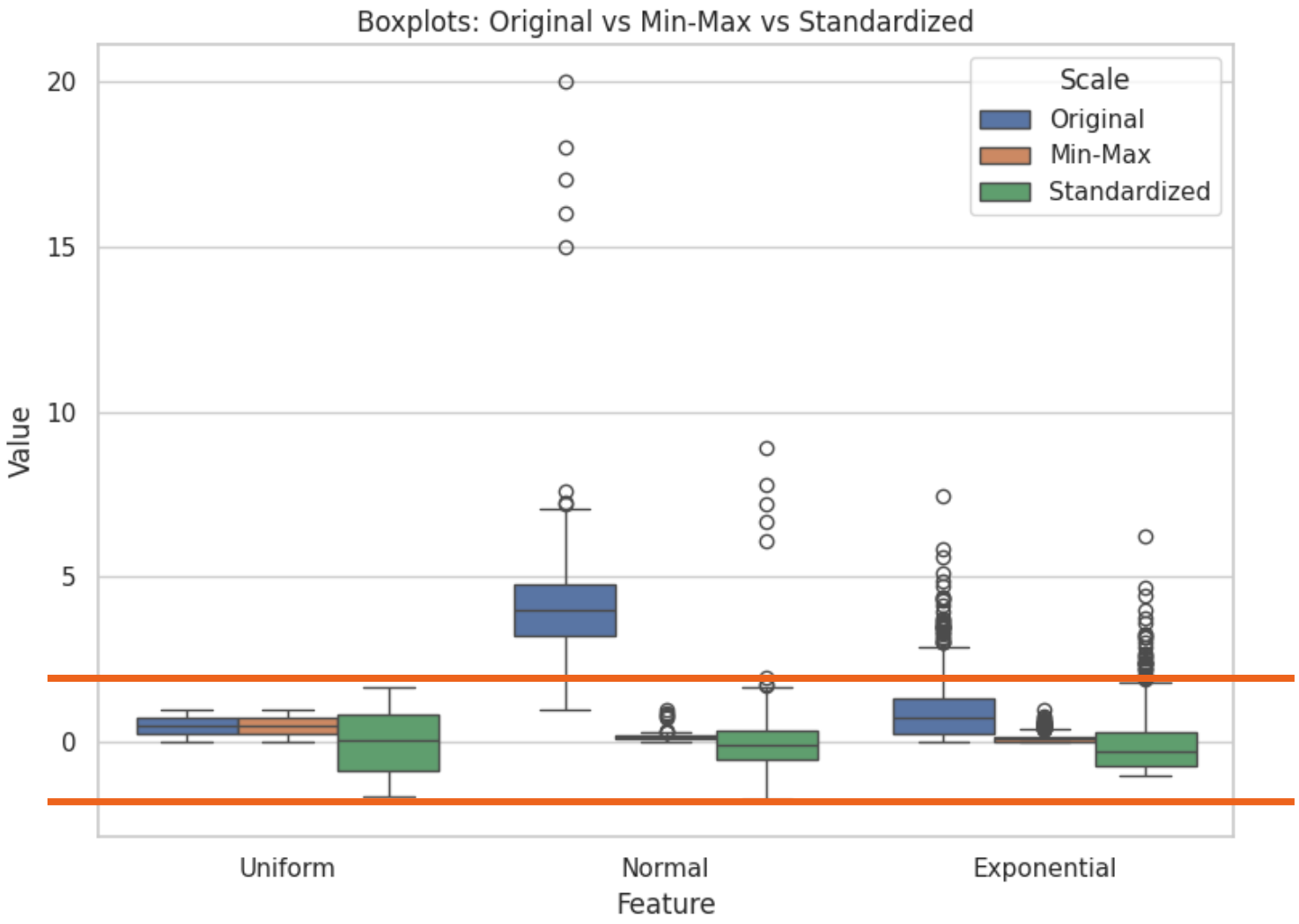
Data normalization



Bringing everything on the same scale
→ fair comparison

Categorical features

Eccentricity	Diameter	Color
0.1	0.66	No
0.3	0.5	No
0.9	1	Yes
0.3	0.33	No
0.3	0.5	No

Ordinal encoding

- Each category label is mapped to an integer
- Implicit meaning of order

Small, medium, large → 0, 1, 2
Dog, cat, cow → 0, 1, 2

One-hot encoding

- Categorical feature is transformed to *n_categories* binary features
- 1 = yes; 0 = no

Red, yellow, blue →

Red	Yellow	Blue
1	0	0
0	1	0
0	0	1

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Categorical features

Eccentricity	Diameter	Color
0.1	0.66	0
0.3	0.5	0
0.9	1	1
0.3	0.33	0
0.3	0.5	0

No = 0
Yes = 1

Ordinal encoding

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How complex should our model be?

DATA **MODEL** **PREDICTION**

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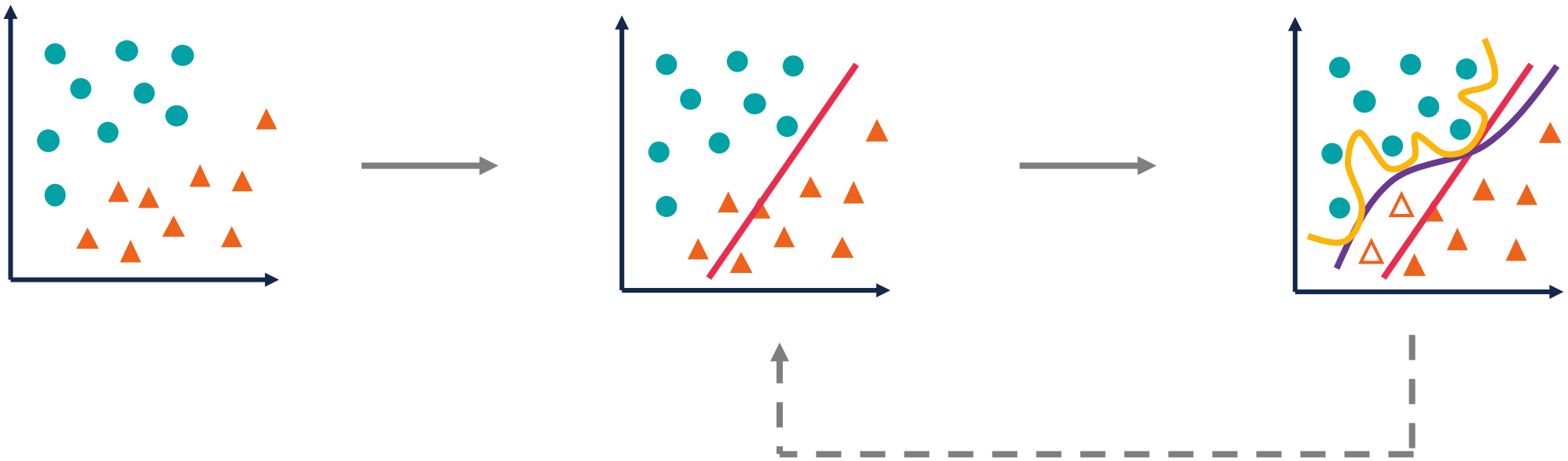

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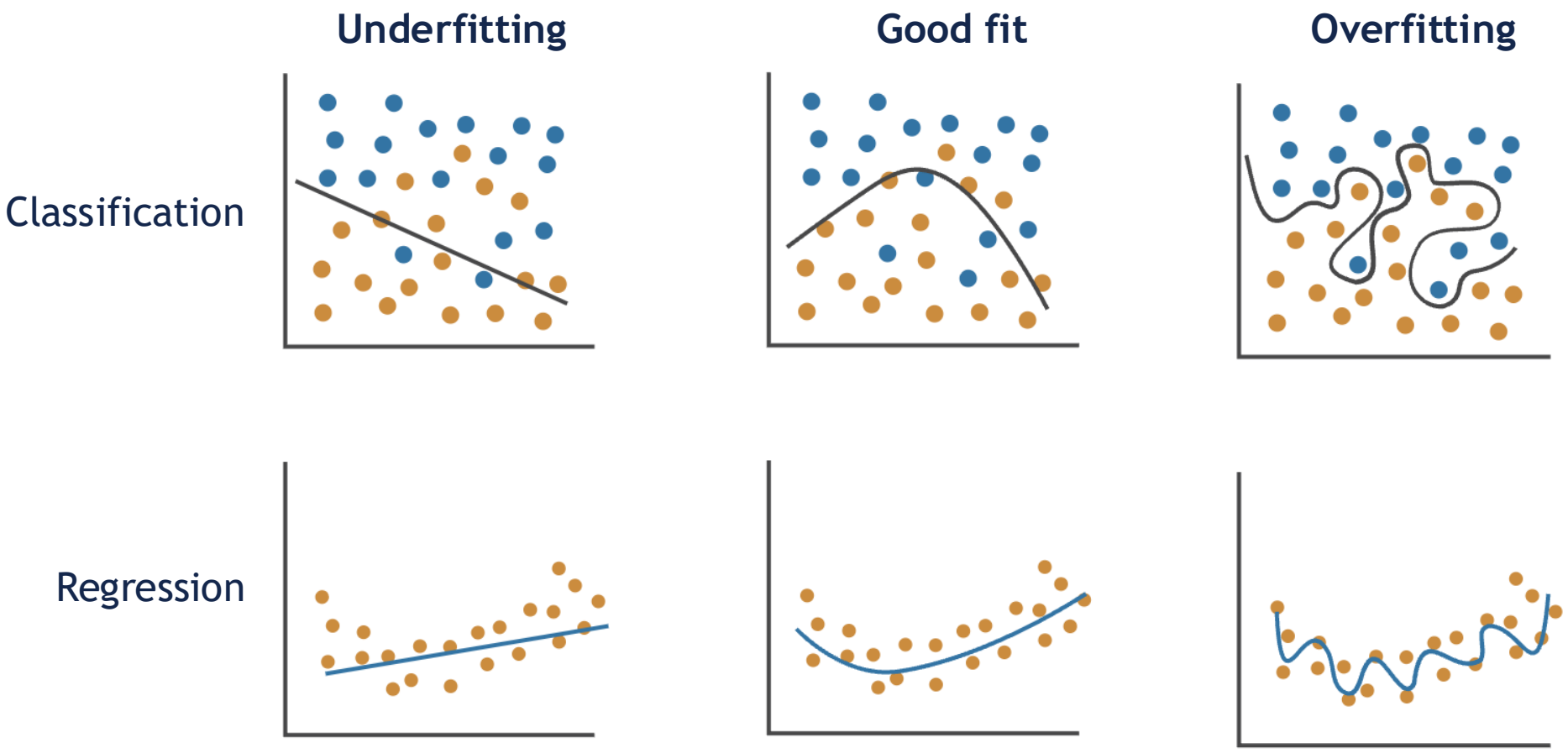
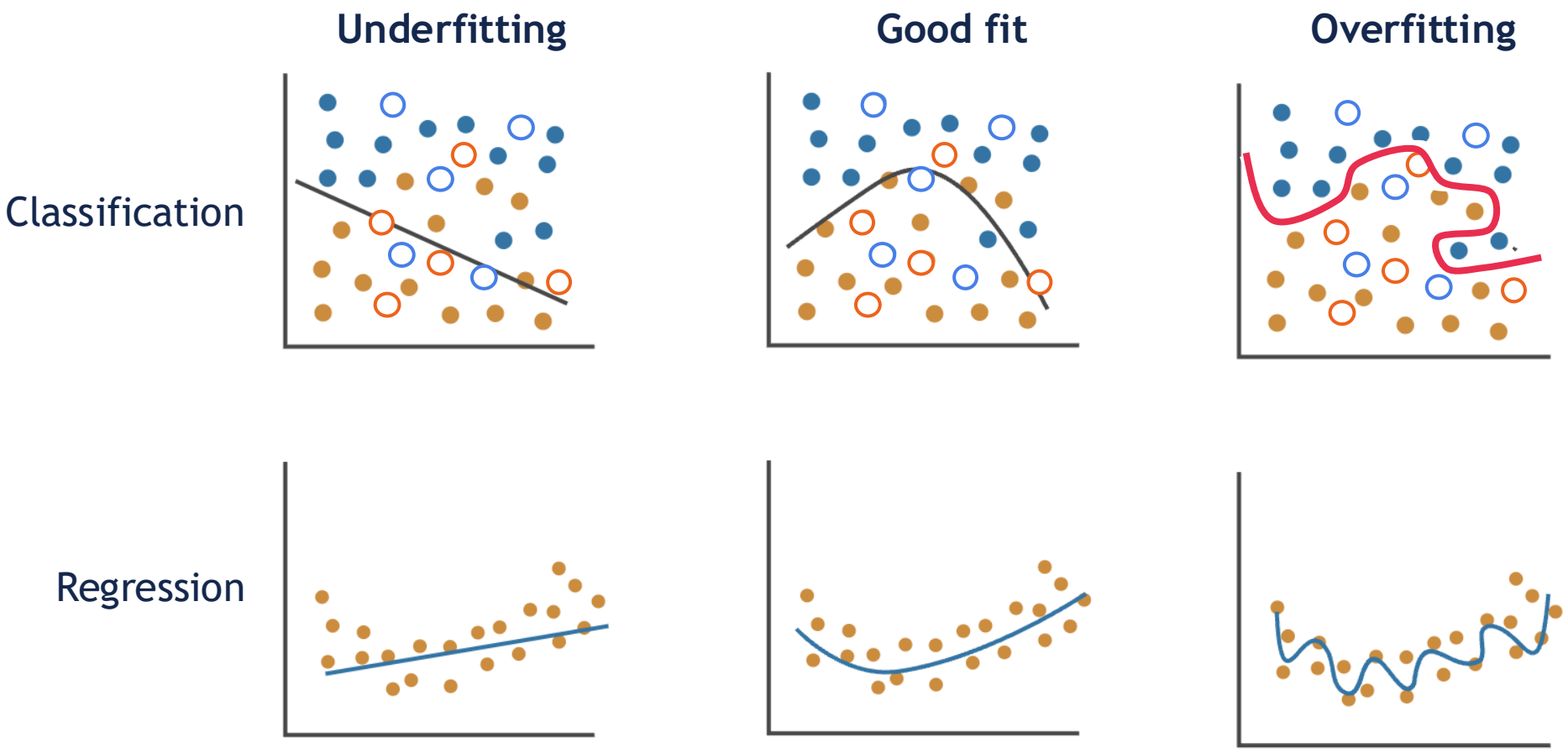
What is the goal of machine learning?

(vs. statistics)

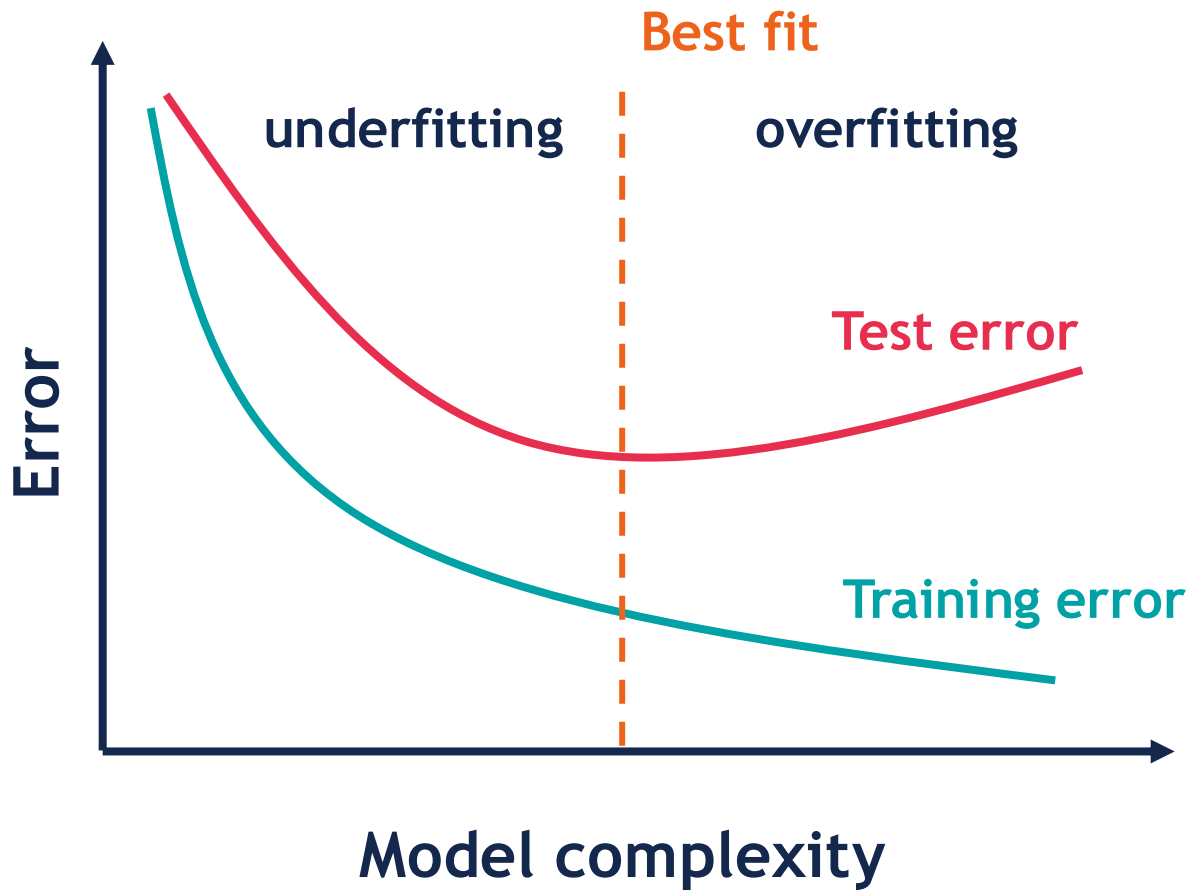
Train-test splitting



Train-test splitting



Train-test splitting



- Test set = **independent dataset** to measure model performance
- Model learns on training data
- Evaluated on (unseen) test data

→ Good generalization to new inputs

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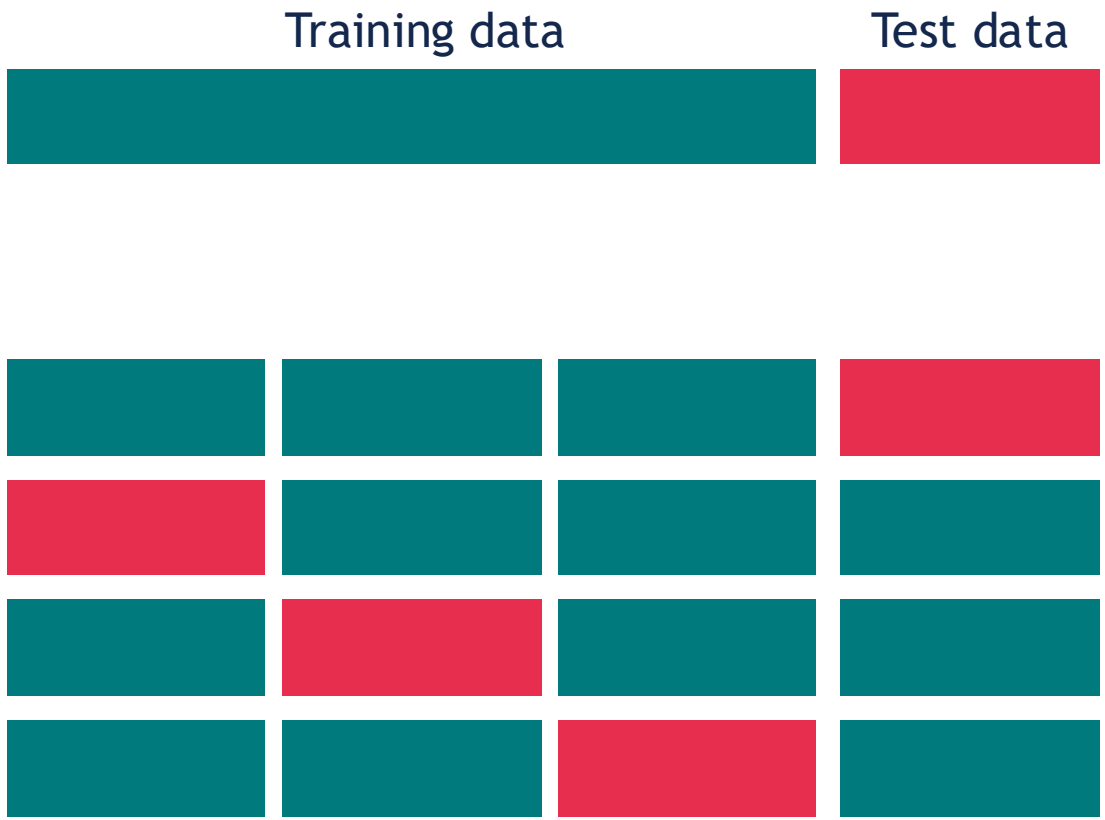
Train-test splitting



Train-test split

- Single split (e.g. 80%/20%)
- Requires more data

Cross-validation

- K splits
- More data efficient
- Reduced bias

Which dataset do we use if we want to …

… **train our model**

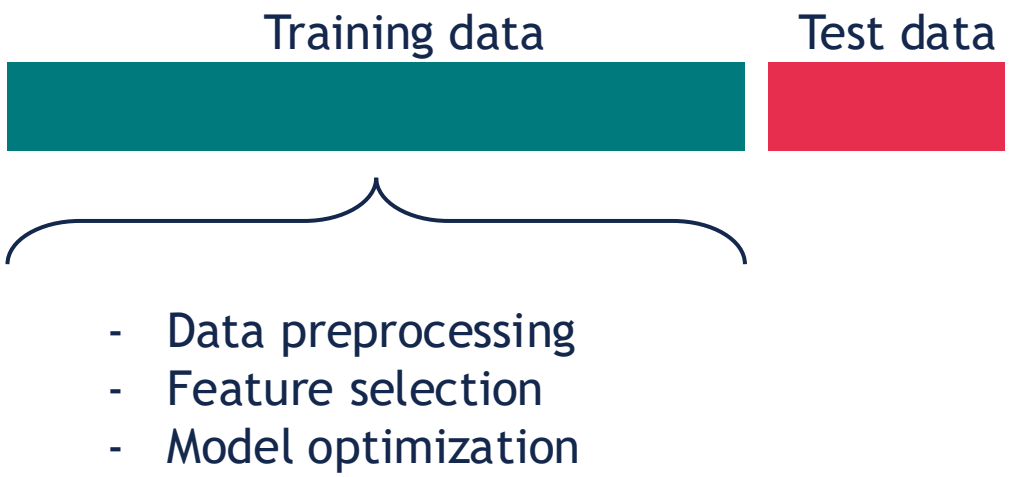
… **select the best features** (e.g. based on a T-test)

… **fit our feature scaler** (i.e. compute μ and σ)

Data leakage

- When **information from the test set influences the model during training**
- Violates the principle of evaluation on **unseen data**
- **Test data "doesn't exist yet"**

→ All fitting should happen **only on the training data**



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